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Stimulating Static Routing Protocol configuration using CISCO Packet Tracer

AIM:-

To stimulate and configure static Routing Protocol using CISCO Packet Tracer.

Static Routing:-

Static Routing is a method of routing in which the routes are manually configured by the network administrator. The administrator specifies the destination network, subnet mask and next-hop address. In static routing, the router does not automatically learn routes from the routers. Therefore, the administrator must configure the required routes manually.

Static routers can also be configured with different administrative Distance. A route with lower administrative distance is preferred over a route with a higher administrative distance.

Procedure:-

(i) Create the required network topology in CISCO packet Tracer.
(ii) Connect the devices using appropriate cables
(iii) Configure the IP address and Subnet masks for all PCs and routers.
(iv) Enable router interfaces and configure static routes in each router.
(v) Configure a primary route with the Lower Administrative Distance

Example

Router(config)# ip route <destination-network>
<subnet mask> <next-hop-IP>

Command Prompt / Router CLI:

Router > enable
Router # configure terminal
Router(config)# ip route 192.168.2.0
255.255.255.0 10.0.0.2

Router (config)# end
Router show ip route

Sample Output

S 192.168.2.0/24 [1/0] via 10.0.0.2

(vi) Configure a backup route with a higher Administrative Distance

(vii) Display the static routing table using,
show ip route static

(viii) Test the connectivity between PC's using pings. Then check path followed by packet using tracer.

Result :-

The static Routing Protocol was simulated and configured successfully using CISCO packet Tracer.